



SQL CLASS NOTE – MODULE 6: ALIAS (AS)

1. Introduction

An **alias** is a temporary name given to a **column** or a **table** in SQL.

- Makes results easier to read.
- Shortens long table/column names.
- Useful when performing calculations or joins.

👉 Think of it like giving someone a **nickname** so they're easier to call.

2. Syntax

Column Alias

```
SELECT column_name AS alias_name  
FROM table_name;
```

Table Alias

```
SELECT t.column_name  
FROM table_name AS t;
```

- AS is optional in most SQL engines.
- The alias lasts only for the duration of the query.

3. Practical Examples with SchoolDB

Example 1: Column alias – full name of student

```
SELECT CONCAT(FirstName, ' ', LastName) AS FullName
FROM Students;
```

Example 2: Column alias – shorten column name

```
SELECT GradeLevel AS Grade, Age AS StudentAge
FROM Students;
```

Example 3: Table alias – shorter name for Students

```
SELECT s.FirstName, s.LastName, s.GradeLevel
FROM Students AS s;
```

Example 4: Table alias in a join

```
SELECT s.FirstName, s.LastName, c.CourseName
FROM Students AS s
JOIN Enrollments AS e ON s.StudentID = e.StudentID
JOIN Courses AS c ON e.CourseID = c.CourseID;
```

📌 Aliases `s`, `e`, and `c` make the query **shorter and cleaner**.

Example 5: Alias in calculations

```
SELECT FirstName, LastName, (Age * 12) AS AgeInMonths
FROM Students;
```

4. Why Use Aliases?

- For **readability**: Makes results look like real-world reports.
- For **simplicity**: Shorter table names = easier to type.
- For **calculated fields**: Rename computed columns.
- For **joins**: Prevents confusion when multiple tables share the same column names.

5. Real-Life Use Cases

- **Schools**: Show `FullName` instead of splitting first and last name.
- **Banks**: Show `AccountBalance` AS `Balance` .
- **Hospitals**: Show `Patients.Name` AS `PatientName` .
- **E-commerce**: Display `Products.Price * Quantity` AS `TotalCost` .

6. Classwork (Practice)

👉 Try these queries on **SchoolDB**:

1. Show all students with a column alias `FullName` .
2. Display students with `Age` as `StudentAge` .
3. Show courses with `CourseName` as `Subject` .
4. Use table alias `s` for `Students` and display their `FirstName` and `GradeLevel` .
5. List students with their age in months (`Age * 12`) as `AgeInMonths` .
6. Use table aliases to show each student's `FullName` and the `CourseName` they are enrolled in.
7. Show all unique grade levels but alias the column as `ClassLevel` .

7. Challenge Question

👉 Write a query that:

- Joins **Students**, **Enrollments**, and **Courses**
- Shows the student's `FullName` (aliased), course `CourseName` (aliased as `Subject`), and enrollment date (aliased as `EnrolledOn`).

✅ That's the **expanded Module 6: ALIAS (AS)**.

Perfect 👍 Let's solve the **Module 6 Classwork + Challenge** step by step.

✅ Solutions to Module 6: ALIAS (AS)

1. Show all students with a column alias `FullName` .

```
SELECT CONCAT(FirstName, ' ', LastName) AS FullName
FROM Students;
```

2. Display students with `Age` as `StudentAge` .

```
SELECT FirstName, LastName, Age AS StudentAge
FROM Students;
```

3. Show courses with `CourseName` as `Subject` .

```
SELECT CourseName AS Subject, Teacher
FROM Courses;
```

4. Use table alias `s` for `Students` and display their `FirstName` and `GradeLevel` .

```
SELECT s.FirstName, s.GradeLevel
FROM Students AS s;
```

5. List students with their age in months ($\text{Age} * 12$) as **AgeInMonths** .

```
SELECT FirstName, LastName, (Age * 12) AS AgeInMonths
FROM Students;
```

6. Use table aliases to show each student's **FullName** and the **CourseName** they are enrolled in.

```
SELECT CONCAT(s.FirstName, ' ', s.LastName) AS FullName,
       c.CourseName AS Subject
FROM Students AS s
JOIN Enrollments AS e ON s.StudentID = e.StudentID
JOIN Courses AS c ON e.CourseID = c.CourseID;
```

7. Show all unique grade levels but alias the column as **ClassLevel** .

```
SELECT DISTINCT GradeLevel AS ClassLevel
FROM Students;
```



Challenge Question

👉 Query that joins **Students**, **Enrollments**, and **Courses**, showing:

- Student's **FullName** (aliased)
- Course **CourseName** (aliased as **Subject**)
- Enrollment date (aliased as **EnrolledOn**)

```
SELECT CONCAT(s.FirstName, ' ', s.LastName) AS FullName,  
       c.CourseName AS Subject,  
       e.EnrollmentDate AS EnrolledOn  
FROM Students AS s  
JOIN Enrollments AS e ON s.StudentID = e.StudentID  
JOIN Courses AS c ON e.CourseID = c.CourseID  
ORDER BY EnrolledOn ASC;
```

📌 This produces a **clean, report-like output** with readable column names.

✅ That wraps up **Module 6: ALIAS**.

👉 Should I move on to **Module 7: LIMIT** next, in the same detailed style (explanations, examples, classwork, and challenge)?